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Bangladesh University of Professionals (BUP)

Department of Computer Science and Engineering (CSE)

Software Test Case Suite

Project: Khet-Khamar

Course Name: CSE 3206: Software Engineering LAB

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Section 1: Post Creation — Test Cases

Feature: Post Creation | Endpoint: POST /api/posts | Project: Khet-Khamar

Tester: _____ Date: _____ Build: _____

Preconditions

- Test user fixture: `{ id:101, role:"user", email:"test@example.com" }`
- Mocks: `db.query`, `media.upload`, `notifications.send`, auth middleware
- Clean all mocks before each test
- Use a valid JWT or mock auth middleware for authenticated requests

Test Cases

ID	Action	Input	Steps	Expected Result
PC-01	Create post	text Content: "Hello World"	1. Tap "Create Post". 2. Type "Hello World". 3. Tap "Publish".	201 — Post appears in feed with correct content.
PC-02	Upload image post	Multipart POST with <code>photo.jpg</code>	1. Tap "Create Post". 2. Tap "Add Photo" & select. 3. Tap "Publish".	201 — Image saved; <code>media_url</code> in DB.
PC-03	Whitespace post	Content: spaces	5 1. Tap "Create Post". 2. Enter 5 blank spaces. 3. Tap "Publish".	400 — Validation error. Post not created.
PC-04	Max length exceeded	Content > 5000 chars	1. Paste text > 5000 chars. 2. Tap "Publish".	400 — “Content too long” error.
PC-05	Unsupported file type	Attach <code>document.exe</code>	1. Tap "Add Photo". 2. Select <code>.exe</code> file. 3. Tap "Publish".	400 — “Invalid file type” error.
PC-06	Photo size exceeded	Attach image > 10 MB	1. Tap "Add Photo". 2. Select 15MB photo. 3. Tap "Publish".	400 — File size error. Upload rejected.
PC-07	Empty submission	No text and no image	1. Tap "Create Post". 2. Leave text/image blank. 3. Tap "Publish".	400 — “Content required” error.

ID	Action	Input	Steps	Expected Result
PC-08	Edit own post	Update text	post 1. Go to own post. 2. Tap "Edit". 3. Change text & Save.	200 — Content updates. "Edited" tag appears.
PC-09	Illegal edit	Edit other's post	1. Intercept edit API. 2. Change target to other's ID. 3. Send payload.	403 — "Access Denied". No changes made.
PC-10	Delete own post	Request delete	1. Go to own post. 2. Tap "Delete". 3. Confirm dialog.	200 — Post removed from feed.
PC-11	Delete other's	Request delete	1. Intercept delete API. 2. Change target to other's ID. 3. Send request.	403 — "Access Denied". Post remains.
PC-12	Post w/o login	POST w/o Auth header	1. Clear session tokens. 2. Attempt post via API.	401 — Redirected to login.
PC-13	XSS Injection	<script>alert(1	1. Enter payload in text. 2. Tap "Publish". 3. Reload feed.	201 — Script sanitized in DB. No alert fires.
PC-14	Rapid duplicate	Same content twice	1. Enter valid text. 2. Double-click "Publish" rapidly.	429/409 — Duplicate blocked.

Section 2: Log In — Test Cases

Feature: Authentication & Access | Endpoint: POST /api/auth/login | Project: Khet-Khamar

Tester: _____ Date: _____ Build: _____

Preconditions

- System contains valid test account: phone 01700000001, password **khetkhamar1**.
- Unregistered test phone number: 018000000099.

Test Cases

ID	Action	Input	Steps	Expected Result
LG-01	Successful login	Valid phone & pass	1. Open App. 2. Enter 01700000001. 3. Enter khetkhamar1 . 4. Tap "Log In".	Redirected to feed. Session token stored.
LG-02	Unregistered phone	Phone 018000000099	1. Enter 018000000099. 2. Enter any password. 3. Tap "Log In".	Error: "No account found with this number."
LG-03	Wrong password	Valid phone & wrong pass	1. Enter 01700000001. 2. Enter incorrect password. 3. Tap "Log In".	Error: "Incorrect password." Field clears.
LG-04	Empty phone	Blank phone & any pass	1. Leave phone blank. 2. Enter password. 3. Tap "Log In".	Validation error: "Phone number is required."
LG-05	Empty password	Valid phone & blank pass	1. Enter valid phone. 2. Leave password blank. 3. Tap "Log In".	Validation error: "Password is required."
LG-06	Invalid phone format	123 in phone field	1. Enter 123 in phone. 2. Enter password. 3. Tap "Log In".	Validation error: "Enter 11-digit mobile number."
LG-07	Account lock-out	5 failed login attempts	1. Enter wrong password 5x. 2. Attempt 6th login.	Error: "Too many attempts. Try in 15 min."

ID	Action	Input	Steps	Expected Result
LG-08	SQL injection	' OR '1'='1 in phone	1. Enter payload in phone. 2. Enter password. 3. Tap "Log In".	Login fails normally. No error trace exposed.
LG-09	Logged-in redirect	Navigate to /login URL	1. Log in successfully. 2. Type /login in URL bar. 3. Press Enter.	Redirected to home/feed. Form not shown.
LG-10	Toggle visibility	Click eye icon on pass	1. Enter password. 2. Tap Eye icon. 3. Tap Eye icon again.	Password reveals on first tap, masks on second.

Section 3: Marketplace — Test Cases

Feature: Marketplace Listings & Interaction | Project: Khet-Khamar

Tester: _____ Date: _____ Build: _____

Preconditions

- Marketplace module active; products present in database.
- Browser/Mobile permissions allow dialer access for “Contact Seller”.

Test Cases

ID	Action	Input	Steps	Expected Result
MP-01	Create Listing	Title, Image=300KB	1. Tap "Sell". 2. Enter "Fresh Rice" & photo. 3. Submit.	Listing saved, visible in feed.
MP-02	Create Listing	Missing Title	1. Tap "Sell". 2. Leave title blank. 3. Submit.	Error “Title is required”, listing not saved.
MP-03	Image Compress	Image=2MB	1. Upload 2MB image. 2. Submit listing.	Auto-compressed ≤ 500KB; listing saved.
MP-04	Language Support	Toggle = Bangla	1. Open app settings. 2. Set lang to Bangla. 3. Open Market.	Fields and labels display in Bangla.
MP-05	Browse Feed	Open as Guest	1. Log out. 2. Navigate to Marketplace.	Ads visible with titles, images, seller info.
MP-06	Filter Ads	Filter by “Seeds”	1. Tap "Filter". 2. Select "Seeds".	Only seed-related ads displayed.
MP-07	Ad Details	Click specific listing	1. Scroll feed. 2. Tap on an ad thumbnail.	Detailed view opens w/ description.

ID	Action	Input	Steps	Expected Result
MP-08	Contact Seller	Click “Contact” button	1. Open ad details. 2. Tap "Contact Seller".	System dialer opens w/ vendor's number.
MP-09	Submit view	Re- Rating=4, Text=Good	1. Go to Seller Profile. 2. Give 4 stars & comment. 3. Submit.	Review saved; avg rating updates.
MP-10	Duplicate view	Re- Review seller same	1. Go to reviewed seller. 2. Attempt second review.	Error: “Already reviewed this seller.”
MP-11	Illegal Edit	Edit another ad	1. Intercept edit API. 2. Swap Ad ID to other's.	Access denied; error message shown.
MP-12	Pagination	Scroll 200+ ads	1. Open Market. 2. Scroll down continuously.	Infinite scroll works without lag.

Section 4: Weather — Test Cases

Feature: Weather Forecasting | Project: Khet-Khamar

Tester: _____ Date: _____ Build: _____

Test Cases

ID	Action	Input	Steps	Expected Result
W-001	Load default	Area: blank	1. Open Weather tab. 2. Ensure location is off.	Defaults to Dhaka weather (14-day).
W-002	Known city	Area: Chit-tagong	1. Tap search bar. 2. Type "Chittagong" & enter.	Displays Chit-tagong weather.
W-003	Mixed case	Area: "dHaKa "	1. Type "dHaKa ". 2. Press Enter.	Recognized as Dhaka.
W-004	Unknown city	Area: Atlantis	1. Type "Atlantis". 2. Press Enter.	Shows "Invalid area" + city list.
W-005	Blank days param	Days: blank	1. API request w/ missing days param.	Shows 14-day forecast.
W-006	Below min days	Days: 3	1. API request w/ days=3.	Shows 7-day forecast.
W-007	Above max days	Days: 30	1. API request w/ days=30.	Shows 21-day forecast.
W-008	Non-numeric days	Days: abc	1. API request w/ days=abc.	Falls back to 14-day forecast.
W-009	Forecast count	Days: 10	1. Request 10 days. 2. Count list items.	Forecast contains exactly 10 entries.
W-010	Current summary	Valid request	1. Load active city. 2. View top banner.	Shows temp, condition, wind, rainfall.
W-011	Day completeness	Valid request	1. Expand a single day.	Shows high/low, condition, rain, wind.

ID	Action	Input	Steps	Expected Result
W-012	Numeric rounding	Valid request	1. Observe temperature values.	Temp/wind shown as whole numbers.
W-013	Unknown condition	Condition unavailable	1. Mock API returning "null".	Displays “Variable”.
W-014	Timestamp	Valid request	1. Check footer of widget.	Readable generated date/time visible.
W-015	Areas list	Valid request	1. Open dropdown list.	Known cities always listed.
W-016	Check ordering	Valid request	1. Observe date sequence.	Sorted chronologically from today.
W-017	API failure	External service down	1. Block external API. 2. Reload page.	Uses fallback values + “Partly Cloudy”.
W-018	Fallback count	API failure	1. Block external API. 2. Count generated items.	Fallback contains exactly 9 days.

Section 5: News — Test Cases

Feature: Agricultural News Feed | Project: Khet-Khamar

Tester: _____ Date: _____ Build: _____

Test Cases

ID	Action	Input	Steps	Expected Result
N-001	External news	API returns 15 items	1. Open News. 2. Count visible articles.	Displays maximum 10 articles.
N-002	API fails, no cache	API fail + no cache	1. Clear cache. 2. Block API & reload.	Loads local database news.
N-003	API fails, cache	API fail + cache exists	1. Preload cache. 2. Block API & reload.	Displays cached news.
N-004	Rate limit retry	Response: “slow down”	1. Mock 429 response. 2. Wait 2s.	Retries and returns news successfully.
N-005	Malformed text	Unexpected text	1. Mock invalid JSON. 2. Reload page.	Graceful fallback without crash.
N-006	Missing title	Title blank	1. Mock blank title field.	Uses “Agricultural update”.
N-007	Weather key-word	“Heavy rain expected”	1. Mock text w/ "rain".	Category mapped = Weather.
N-008	Seed keyword	“New rice seed”	1. Mock text w/ "seed".	Category mapped = Seeds.
N-009	Pest keyword	“Pest attack alert”	1. Mock text w/ "pest".	Category mapped = Pest Control.
N-010	Irrigation map	“Efficient irrigation”	1. Mock text w/ "irrigation".	Category mapped = Irrigation.
N-011	Technology map	“AI farming tools”	1. Mock text w/ "AI".	Category mapped = Technology.
N-012	Government map	“Subsidy for farmers”	1. Mock text w/ "Subsidy".	Category mapped = Government.

ID	Action	Input	Steps	Expected Result
N-013	Tips keyword	“Tips for crop growth”	1. Mock text w/ "Tips".	Category mapped = Tips.
N-014	No keyword match	Generic article	1. Mock generic text.	Category mapped = Market.
N-015	Featured articles	External list	1. Look at top of feed.	First 2 marked featured.
N-016	Article w/ image	Image present	1. Feed loads w/ images.	Image displayed properly.
N-017	Article w/o image	No image provided	1. Mock missing image url.	Image field empty/null.
N-018	Missing source	Source blank	1. Mock blank source.	Uses “External Source”.
N-019	Verify timestamp	Valid article	1. Check article metadata.	Created/seen time shown.
N-020	Article w/ URL	Link available	1. Click "Read More".	Redirects to URL.
N-021	Article w/o URL	Link missing	1. Mock blank link.	URL field empty / hidden.
N-022	Max display	20 articles fetched	1. Scroll to bottom.	Only 10 displayed.
N-023	Share w/o title	Title blank	1. Tap Share. 2. Clear title & submit.	Error: title required.
N-024	Share w/ spaces	“ Rice News ”	1. Share w/ space padding.	Spaces trimmed before saving.
N-025	Optional fields	Only title provided	1. Tap Share. 2. Fill only title.	Share succeeds.
N-026	Successful share	Valid logged-in	1. Log in. 2. Share article.	Confirmation + generated ID.
N-027	Share logged out	Valid submission	1. Log out. 2. Share article.	Login-required error.
N-028	Shared word	“Best seed options”	1. Share text w/ "seed".	Category mapped = Seeds.

ID	Action	Input	Steps	Expected Result
N-029	Missing fields	Null fields	1. Mock null responses.	No crash, handled gracefully.
